

# Raphaël URFIN

Centre de Sciences des Données, 45 rue d'Ulm – 75005 Paris – France

## About me

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I am a first year PhD student at École Normale Supérieure - PSL in Paris supervised by Giulio Biroli (ENS) and Marc Mézard (Bocconi University) working on the theory of diffusion models. My background is in theoretical physics.

## Education

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- **PhD in Physics** **ENS-PSL, Paris**  
*Subject: Generative Diffusion and Statistical Physics*  
LPENS & Centre de Sciences des Données  
Supervisors: Giulio Biroli (ENS) and Marc Mézard (Bocconi University)  
2025 - 2028(exp.)
- **ENS Diploma** **ENS-PSL, Paris**  
*Interdisciplinary Diploma validating my studies at ENS.*  
2021-2025
- **M2 ICFP Theoretical Physics Track** **ENS-PSL, Paris**  
*First Class Honors, 17.38/20*  
2023 - 2024  
Relevant classes: Advanced Statistical Physics, Disordered Systems, Machine Learning.
- **M1 ICFP** **ENS-PSL, Paris**  
*First Class Honors, 17.8/20*  
2022 - 2023  
Relevant classes: Phase Transitions, Introduction to Quantum Field Theory, General Relativity.
- **Bachelor in Physics and Mathematics** **ENS-PSL, Paris**  
*First Class Honors, 17.38/20 (Physics) and 16.36/20 (Mathematics)*  
2021 - 2022

## Publications

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\* denotes equal contribution.

- Tony Bonnaire\*, **Raphaël Urfin**\*, Giulio Biroli, Marc Mézard.  
*Why Diffusion Models Don't Memorize: The Role of Implicit Dynamical Regularization in Training.*  
NeurIPS 2025, **Best Paper Award (4 papers among +20000 submissions)**  
<https://openreview.net/forum?id=BSZqpqqqM0>

## Conferences, Workshops and Seminars

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### Invited Talks.....

- **LemanTH 2026** **Lausanne, Switzerland**  
*Workshop*  
April 2026
  - Talk on the Memorization/Generalization transition in Diffusion Models.
- **Vitelli's Group Meeting** **Online**  
*Group meeting*  
March 2026
  - Talk on the non-reciprocal p-spins.
- **Scattering Club** **Online**  
*Invited Seminar*  
February 2026
  - 1h invited talk on the Memorization/Generalization transition in diffusion models. Webinar organized by the data science for astrophysics group of LPENS.

- **Voxel51 Best of NeurIPS 2025** **Online**  
*Invited Seminar* *January 2026*
  - 30 min invited talk on the Memorization/Generalization transition in diffusion models.
- **Centre de Sciences des Données Internal Seminar** **Paris, France**  
*Invited Seminar* *December 2025*
  - 1h invited talk on the Memorization/Generalization transition in diffusion models.
- **Machine Learning & Signal Processing @ ENS Lyon** **Lyon, France**  
*Invited Seminar* *November 2025*
  - 1 hour invited talk on the Memorization/Generalization transition in diffusion models.

## Contributed Talks.....

- **Journées de Physique Statistique 2026** **Paris, France**  
*Conference* *January 2026*
  - 4-minute flash talk on the Memorization/Generalization transition in Diffusion Models.
- **EurIPS 2025** **Copenhagen, Denmark**  
*Conference* *December 2025*
  - Poster+Oral Presentation of our paper accepted at NeurIPS 2025.
- **NeurIPS in Paris 2025** **Paris, France**  
*Workshop* *November 2025*
  - Poster+Oral Presentation of our paper accepted at NeurIPS 2025.
- **StatPhys 29** **Florence, Italy**  
*Conference* *July 2025*
  - 15-minute contributed talk on the Memorization/Generalization transition in diffusion models.
- **Youth in High Dimensions** **Trieste, Italy**  
*Workshop* *July 2025*
  - 15-minute contributed talk on the Memorization/Generalization transition in diffusion models.
- **Journées de Physique Statistique 2025** **Paris, France**  
*Conference* *January 2025*
  - 4-minute flash talk presenting results from my M2 internship.

## Summer Schools.....

- **Beg Rohu Summer School of Statistical Physics** **Beg Rohu, France**  
*Summer School* *June 2025 & June 2026*
  - **2026 Edition (Optimization and Control of Nonequilibrium Systems):** Lectures by international researchers (e.g. G. Peyré, G. Rotskoff ...). Presented an updated poster on Memorization/Generalization in diffusion models.
  - **2025 Edition (Learning with Machines, Physics and Mind):** Lectures on Machine Learning and Statistical Physics by international researchers (e.g., Yann LeCun, Julia Kempe, Stéphane Mallat, Marc Mézard ...). Presented initial poster on Memorization/Generalization in diffusion models.
- **Complex and Glassy Systems** **Cargese, France**  
*Summer School* *July 2024*
  - Lectures on Statistical Physics and interdisciplinary applications by international researchers (e.g. Marc Mézard, Eric Vanden-Eijnden, Valentina Ros, Guy Bunin).

## Awards and Grants

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- 2025: Recipient of one of the four best paper awards at NeurIPS 2025. **(4 papers among +20000 submissions)**
- 2025: CDSN, Competitive PhD scholarship for ENS students

## Reviewing

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- Machine Learning Workshops: EurIPS 2025 Workshop Principles of Generative Modeling (PriGM), ICLR 2026 Workshop DeLTa, ICLR 2026 Workshop Sci4DL.

- Machine Learning Journals: TMLR

## Skills

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- Languages: French (Native), English (Fluent), Italian (Fluent).
- Software skills: Python (Pytorch), Matlab, Latex.

## Teaching Experience

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- TA Machine Learning (M2, Master ICFP, ENS-PSL, 11h), class taught in English with Marc Lelarge and Leonardo Defilippis. Spring 2026.
- TA Mathematics Tutoring for first year student in Physics at ENS. 36h. class taught in French with Amir-Kian Kashani-Poor. Fall 2025
- TA Stochastic Processes for physics (M1 level, Master ICFP, ENS-PSL), 16 hours, class taught in English with Marylou Gabrié. Fall 2025
- "Khôlles" (Preparation for the oral Exam for French "Grandes Écoles") in Mathematics and Physics for Classes Préparatoires PC, Lycée Stanislas (2022–2024)

## Relevant Experience

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- **Statistical Physics and Diffusion Models** **Bocconi University, Milano, Italy**  
*Research Internship* *February 2025-May 2025*
  - Supervisor: Marc Mézard, Departement of Computing Sciences.
  - Subject: The Memorization/Generalization transition in diffusion models.
  - Resulted in a publication accepted at Neurips 2025
- **Statistical Physics and Diffusion Models** **École Normale Supérieure-PSL, France**  
*Research Internship* *April 2024-January 2025*
  - Supervisor: Giulio Biroli, Centre de Sciences des Données.
  - Subject: The effect of implicit regularization in diffusion models.
- **Emergent behaviors in large ecosystems** **University of Cambridge, UK**  
*Research Internship* *January-July 2023*
  - Supervisor: Camille Scalliet, Soft Matter Group, Department of Applied Mathematics and Theoretical Physics.
  - Using tools from disordered systems (Cavity Method, Random Matrix Theory...) to understand emergent collective behaviors in ecological systems.